

Article

A Coordinated Restoration Scheduling Strategy for Distribution Network Sources Under Typhoon Weather Considering Correlation Effects

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Abstract

To mitigate large-scale blackout risks in urban distribution systems under typhoon-induced extreme weather and to reduce post-disaster restoration costs, this study proposes a resilience-oriented spatiotemporal co-optimization framework integrating transportation networks, power grids, and distributed energy resources. First, a city-scale typhoon spatiotemporal model is established, integrating static wind field, dynamic evolution, and trajectory-based mobility with urban-geometry-driven wind speed correction to characterize the spatiotemporal progression of extreme wind hazards. Second, the time-varying failure rates of distribution network components are quantified by explicitly accounting for network topology correlations, while the spatiotemporal dispatchability and output characteristics of distributed resources under disaster conditions are systematically modeled. Third, a pre-disaster proactive deployment model is formulated to minimize load curtailment costs and resource allocation expenditures. The model integrates active network reconfiguration with coordinated placement of distributed generation (DG) and mobile energy storage systems (MESSs), enabling resilience-enhancing pre-positioning strategies. Subsequently, a post-disaster restoration scheduling model is developed with the objective of minimizing unserved load. By embedding traffic flow constraints and optimal path computation under disrupted transportation conditions, the proposed framework realizes spatiotemporal coordination among MESSs, DG, and electric vehicles (EVs), thereby accelerating system-level recovery. Finally, the effectiveness of the proposed strategy is validated on a 51-node urban distribution system located in eastern coastal China, demonstrating significant improvements in restoration performance and resilience enhancement.



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1. Introduction

In recent years, the increasing frequency and intensity of extreme natural disasters—such as typhoons, floods, and severe winter storms—driven by global climate change have posed unprecedented challenges to the secure operation of power systems, particularly distribution networks. As the “last mile” of electricity delivery, distribution systems feature geographically dispersed infrastructure and complex network structures

yet often exhibit relatively limited resilience against extreme events. Consequently, they are highly vulnerable to large-scale outages during severe disasters, which can lead to substantial socio-economic losses. Against this backdrop, enhancing the restoration capability of distribution networks has emerged as a critical research priority in modern power system studies [1]. System resilience extends beyond the ability to withstand disruptive events; it also encompasses the capacity to rapidly restore electricity supply following disturbances. Therefore, restoration capability has become a fundamental metric for evaluating the overall performance of distribution networks in responding to extreme events [2].

Throughout the lifecycle of a typhoon—from formation and propagation to landfall and eventual dissipation—the storm’s trajectory, wind-field structure, and intensity exhibit pronounced temporal dynamics and spatial heterogeneity [3]. Consequently, distribution network components located at different nodes and lines are subjected to significantly different wind speed loads at different time periods [4]. Existing studies have attempted to characterize typhoon wind fields using various modeling approaches. For instance, Batts-based formulations have been employed to construct static wind-field representations, where the storm center location and track parameters are used to describe sinusoidal movement trajectories [5]. An improved wind-field model derived from the Holland formulation was proposed in [6], in which the affected region at a given time is approximated by two concentric circles. In [7], the post-landfall wind-field evolution is described using parameters such as landfall angle, pressure difference, and temporal progression. Meanwhile, in [8], directly partitions typhoon-affected regions according to landfall trajectory, landfall radius, and duration, combined with wind speed variations. Despite these efforts, most existing studies assume that key wind-field parameters remain constant within the analyzed time horizon—particularly after landfall—making it difficult to capture critical dynamic characteristics such as wind-field expansion, intensity attenuation, and trajectory perturbations during typhoon evolution. Furthermore, with respect to wind-speed correction, most studies rely primarily on surface roughness coefficients or empirical correction factors [9], without systematically incorporating the influence of urban morphological features, such as building height and building density, on near-surface wind fields. As a result, the wind-speed estimation at distribution network nodes often remains overly simplified. Overall, current typhoon modeling approaches still exhibit limitations in representing spatiotemporal evolution processes, capturing urban environmental effects, and coupling with the actual geographic information of distribution network infrastructure. Therefore, it is necessary to develop an integrated typhoon spatiotemporal model that combines static wind-field representation, dynamic evolution mechanisms, and trajectory-based mobility characteristics [10], thereby providing a more realistic hazard input for subsequent failure probability modeling of distribution network components. In summary, existing typhoon models still lack an integrated framework that simultaneously captures spatiotemporal evolution processes, urban canopy effects, and coupling with actual distribution network geographic information, making it difficult to provide sufficiently accurate hazard input for subsequent component-level failure probability modeling.

Under typhoon conditions, failures in distribution network infrastructure fundamentally originate from the structural impacts of strong wind loads on critical components such as conductors and utility poles [11]. Existing studies on equipment failures in distribution systems during typhoon events generally take wind speed as the primary driving factor, establishing probabilistic mappings between wind speed and the failure likelihood of conductors or poles to quantify equipment outage risks [12]. For example, the failure probability of distribution lines has been estimated by comparing calculated conductor wind loads with their maximum wind-bearing capacity [13]. In [14], the distribution network is discretized into spatial grids, within which conductors are assumed to share identical

failure probabilities, thereby constructing a one-to-one mapping between wind speed and conductor failure probability. Similarly, ref. [15] develops a failure rate calculation and correction model for utility poles to represent the outage risk of distribution infrastructure under disaster conditions. Although these studies recognize conductors and poles as the primary vulnerable components, they typically approximate the failure probability of an entire distribution line using a single component-level probability. Such simplifications overlook the structural fact that distribution lines are composed of conductors and poles connected in series, where the failure of any component may lead to line outages. Moreover, the modeling of failure mechanisms for conductors and poles remains relatively simplified in existing work. For instance, when calculating wind loads on conductors, the influence of the angle between wind direction and conductor orientation is often neglected [16]. Likewise, in pole stress assessments, a uniform wind speed parameter is frequently adopted without considering variations in equivalent wind speed caused by structural characteristics such as pole height, resulting in an insufficiently detailed representation of equipment vulnerability [17]. After estimating the failure probabilities of individual distribution lines, most studies directly treat them as the final outage probabilities [18] for network-level analysis, implicitly assuming independence among different lines. However, under extreme weather conditions such as typhoons, the dense spatial distribution of poles and the close proximity of distribution lines mean that the failure of local infrastructure can significantly increase the damage risk of neighboring components. As a result, failures in distribution networks often exhibit pronounced spatial correlation and cascading characteristics [19]. Neglecting such network-level dependencies may lead to an underestimation of failure scales in locally high-risk regions, thereby affecting the accurate assessment of overall disaster impacts on distribution systems.

Significant progress has been made in recent years in leveraging distributed energy resources for post-disaster restoration of distribution systems. Existing studies primarily focus on post-event recovery strategies, including network reconfiguration [20], islanding operation [21], and the coordinated dispatch of distributed generation and energy storage resources [22], which can improve load restoration performance to a certain extent. For instance, ref. [23] establishes a post-disaster restoration framework by optimizing the siting and capacity allocation of distributed generation units within distribution networks. Similarly, ref. [24] proposes a restoration strategy that dynamically integrates island partitioning with the spatiotemporal scheduling of emergency power supply vehicles. While these studies provide valuable insights into post-disaster recovery operations, they largely concentrate on the restoration stage itself and fail to fully exploit the potential of proactive pre-disaster decision-making to reduce system vulnerability. Furthermore, several studies [25–27] employ mobile power sources or repair crews as single or fixed combinations of resources to support restoration efforts. However, the diversified participation of distributed energy resources in enhancing distribution system resilience has received relatively limited attention, which may constrain the efficiency and economic performance of restoration processes. More recently, studies such as [28,29] have explored the use of mobile energy storage systems and electric vehicles to improve post-disaster resilience. Nevertheless, important issues remain insufficiently addressed, including optimal routing of mobile resources after disasters, travel behavior and scheduling constraints on the user side, and traffic flow prediction under disrupted transportation conditions. Overall, existing research on distribution system restoration still exhibits several limitations, particularly in terms of proactive pre-disaster preparedness, multi-timescale restoration planning, coordinated optimization of heterogeneous distributed resources, and the prediction of post-disaster operating environments. Therefore, it is necessary to develop an integrated optimization framework that simultaneously considers both pre-disaster preparedness and post-disaster

restoration, incorporates multiple types of flexible resources, and captures the dynamic evolution of operating conditions under disaster scenarios, thereby enabling a systematic enhancement of distribution system resilience.

Beyond distribution network resilience, the role of transportation networks in post-disaster restoration has attracted increasing attention. The traffic capacity of post-disaster road networks directly determines the dispatch efficiency of mobile emergency resources (e.g., mobile energy storage vehicles, repair crews). Several studies have introduced traffic flow assignment theory into post-disaster restoration problems, establishing optimal routing models under road congestion conditions [30,31]. However, these studies mostly treat the transportation network and the distribution network as independent systems, overlooking their dynamic interactions during post-disaster restoration. In recent years, interdependent network theory has provided theoretical tools for modeling coupled power-transportation systems [32]: for example, Bahmani, M.H. proposed a resilience assessment framework for interdependent transportation-power systems, using Markov chain models to characterize the propagation effects of disasters between the two networks [33]; Candan, A.K. investigated the post-disaster dispatch potential of electric vehicles as coupling nodes between the two networks [34]. Nonetheless, existing interdependent network studies are mostly based on static or quasi-static coupling assumptions and lack dynamic coordination analysis of the two networks under a unified driving force such as the spatiotemporal evolution of extreme meteorological events like typhoons [35]. This constitutes a key research motivation for the present study.

To address the aforementioned challenges, this paper proposes a coordinated restoration scheduling strategy for distribution systems that integrates transportation networks, power grids, and distributed energy resources.

First, a spherical coordinate system is established based on the fundamental parameters and geographic information of the distribution network. Using real-time meteorological information, a comprehensive typhoon model is developed to simulate the storm trajectory and wind-field intensity evolution while incorporating the influence of urban geographical environments during landfall.

Second, the failure mechanisms of distribution network infrastructure under typhoon conditions are systematically analyzed. A distribution network outage model is constructed by jointly evaluating the structural strength of poles and conductors. Meanwhile, the correlation among network component failures is explicitly considered, enabling the derivation of time-varying failure probabilities for distribution network components during typhoon events.

Finally, the proposed framework jointly considers restoration efficiency at distribution network nodes and the deployment cost of heterogeneous distributed resources. In the pre-disaster deployment stage, the variability of wind and photovoltaic generation is taken into account, together with potential network reconfiguration scenarios during typhoon events. With the objective of minimizing the comprehensive deployment cost of distributed resources, the optimal siting of energy storage systems and the operating schedules of wind and photovoltaic units are determined. In the post-disaster restoration stage, coordinated operation among multiple energy resources and the dynamic dispatch of mobile energy storage systems are implemented to minimize load curtailment costs, thereby enhancing the overall resilience of the distribution network.

2. Typhoon Spatiotemporal Model

2.1. Static Wind Field Model

Existing studies on typhoon modeling indicate that the characterization of typhoon wind fields can be interpreted as the combined effect of three key components: a static

wind-field structure, wind-field evolution dynamics, and the storm trajectory. With respect to the static wind field, an improved Holland wind-field model is adopted in this study. Previous research suggests that the spatial influence of a typhoon can be approximately represented by two concentric circular regions. Within the affected area, the wind speed v at any location is expressed as a piecewise function of the distance d between that location and the typhoon eye. Outside the typhoon influence region, the wind speed is assumed to be zero. For locations within the affected region, the wind speed exhibits a distinct radial distribution pattern: as the distance d increases from the typhoon eye to the radius of maximum wind speed, the wind speed increases monotonically; subsequently, as d increases from the radius of maximum wind speed to the outer radius of typhoon influence, the wind speed decreases monotonically.

$$\begin{cases} v = K \cdot v_m \cdot \left(1 - \exp\left(-\frac{d}{r_m} \cdot \ln\left(\frac{K}{K-1}\right)\right)\right) & 0 \leq d \leq r_m \\ v = v_m \cdot \exp\left(-\frac{\ln(\beta)}{r-r_m} \cdot (d - r_m)\right) & r_m \leq d \leq r \\ v = 0 & d \geq r \end{cases} \quad (1)$$

where K denotes an empirical parameter; v_m represents the maximum wind speed within the typhoon static wind field; r_m is the radius corresponding to the maximum wind speed; r denotes the radius of the typhoon influence region; and β is the typhoon boundary factor, defined as the ratio between the maximum gradient wind speed and the boundary wind speed.

The empirical coefficient K and the boundary factor β are used to characterize the radial wind-speed distribution of the improved Holland-type wind field. Their baseline values are selected according to commonly used empirical typhoon wind-field formulations and can be calibrated using local meteorological-station wind-speed records when such data are available.

In Equation (1), the theoretical jump of the third branch at $d = r$ is physically replaced by a continuous transition; in practical computations, $v(d) = \max(v(d), 0)$ is adopted. The parameter $\beta \in [1.2, 1.8]$ is the wind field shape parameter, whose typical values are determined by empirical fitting to measured typhoon wind profiles.

By comparing the distance between each distribution network node and the center of the typhoon wind field at the current time step, the wind speed at the corresponding node can be determined. The distance calculation is performed using a spherical distance formula.

$$r = R_w \arccos[\sin y_1 \sin y_2 + \cos y_1 \cos y_2 \cos(x_1 - x_2)] \quad (2)$$

where R_w denotes the earth's radius, generally taken as 6371 km, the coordinates (x_1, y_1) and (x_2, y_2) represent the latitude and longitude of the node and the center of the wind circle, respectively.

2.2. Dynamic Evolution Model

After a typhoon makes landfall from the ocean into urban areas, its wind speed is influenced by urban environmental factors such as building height and surface roughness. As a result, the radii of both the inner and outer wind fields expand according to a certain proportional coefficient, while the storm intensity gradually weakens. To achieve a more accurate representation of typhoon evolution after landfall, the landfall time and the wind-field radius are selected as key input features of the dynamic wind-field evolution

model. Based on this principle, the mathematical formulation of the post-landfall typhoon wind-field evolution model is given as follows.

$$\begin{cases} r_t = r_{t0} + \lambda t^2 \\ r_{mt} = r_{mt0} + \lambda t^2 \end{cases} \quad (3)$$

$$V_{\max}(t) = V_{\max}(0) \cdot \exp\left(-\frac{t}{\tau_d}\right)$$

where λ denotes the expansion coefficient of the typhoon wind-field radius after landfall. According to the equation, the wind-field radius increases quadratically with time. where $V_{\max}(t)$ is the central maximum wind speed (m/s) at time t after landfall, $V_{\max}(0)$ is the central maximum wind speed at the time of landfall, and τ_d is the intensity decay time constant, reflecting the decay rate of the typhoon central pressure difference, whose value is determined from historical data or meteorological forecasts. Solving Equations (3) and the supplementary equations below simultaneously jointly yields the wind field intensity and extent within the typhoon-affected region at any given time.

Under strong wind conditions, it is difficult to directly measure the actual wind speed acting on each component within the geographic environment of a distribution network. Moreover, wind speed data obtained from nearby meteorological stations cannot accurately reflect the local wind conditions at specific network nodes. Therefore, it is necessary to apply wind speed corrections to estimate the actual wind speed at each node. According to meteorological studies, the wind speed at a given location is closely related to factors such as relative elevation and surface roughness. In addition, when a typhoon makes landfall and passes through urban areas, the wind speed may vary significantly due to the influence of buildings with different heights, which alter the near-surface wind field.

Let the change in wind speed caused by roughness variation be Δv_z , and the calculation formula is as follows

$$\Delta v_z = \left\{ \left[\frac{\ln(\frac{h}{z_1})}{\ln(\frac{\delta}{z_0})} \right] \left[\frac{\ln(\frac{\delta}{z_1})}{\ln(\frac{h}{z_0})} \right] - 1 \right\} \times v_0 \quad (4)$$

where δ denotes the height of the internal boundary layer, which is typically taken as 800 for urban distribution networks; z_0 represents the reference surface roughness length, generally set to 1.0; z_1 denotes the surface roughness length corresponding to the location of different distribution network nodes; and h is the reference wind measurement height, which is usually taken as 10 m. The roughness coefficients in Table 1 are selected from standard terrain-category values in boundary-layer meteorology and wind engineering. For each distribution-network node, the coefficient is assigned according to the dominant underlying surface type around the node.

Table 1. Reference values for different surface roughness.

Underlying Surface Type	Roughness
Forest	0.05~0.10
Low-rise sparsely populated suburbs	0.2~0.4
A district of medium-rise buildings	0.4~0.7
Central Business district	1.0~1.5
Flat ground	0.021
Lake	0.0002~0.0003

The ground roughness in the formula can refer to the following Table 1.

To resolve the issue of an undefined logarithmic term within the urban canopy, a street-canyon reference height z_{ref} is defined as a reference plane above the canopy. Based

on wind profile theory, $z_{ref} = \max(2.5H, h + 10)$ is adopted to ensure $z_{ref} > d$ strictly holds for all urban morphologies. The typical effective height of distribution network conductors and poles is denoted by h .

The roughness length z_0 is parameterized using the Macdonald scheme, which explicitly incorporates the building density λ :

$$z_0 = \left(1 - \frac{d}{H}\right) \exp\left(-\left[0.5\beta \frac{C_D}{\kappa^2} \left(1 - \frac{d}{H}\right) \lambda_f\right]^{-0.5}\right) \times H$$

where λ_f is the frontal area ratio (proportional to building density λ , C_D is the drag coefficient (taken as 1.2), $\kappa = 0.4$ is the von Kármán constant, and β is an empirical constant.

Based on the above definitions, the wind speed correction term Δv_h at the distribution equipment height h is expressed as the difference between the reference wind speed $V(h)$ and the open-terrain wind speed v_0 :

$$\begin{aligned} \Delta V_h &= V(h) - V_0 = V_0 \cdot \frac{\ln\left(\frac{h-d}{z_0}\right)}{\ln\left(\frac{z_{ref}-d}{z_0}\right)} - V_0 \\ &= V_0 \left[\frac{\ln\left(\frac{h-d}{z_0}\right)}{\ln\left(\frac{z_{ref}-d}{z_0}\right)} - 1 \right] \end{aligned} \quad (5)$$

Analysis of the above equation reveals: (i) When h approaches d , the numerator approaches $-\infty$ (a lower bound is set in practical calculations), yielding $\Delta V_h \rightarrow V_0$, indicating very low wind speed inside the canopy. (ii) When $h = z_{ref}$, $\Delta V_h = 0$, and the correction term vanishes, consistent with its physical role. (iii) When $h > z_{ref}$, the correction term becomes positive, reflecting the characteristic of increasing wind speed at higher altitudes. Furthermore, the building density λ indirectly but explicitly enters the correction formula through its influence on d and z_0 .

The building-density parameter λ is defined as the local building coverage ratio of the distribution-network node, ranging from 0 to 1. When high-resolution building-footprint data are unavailable, λ is assigned according to the corresponding urban morphology category in Table 2. Therefore, by considering the effects of surface roughness as well as the height and density of urban buildings, and by applying corrections to both the lateral and longitudinal wind speed components, the corrected wind speed at urban distribution network nodes can be accurately obtained.

$$v_i = v + \Delta v_z + \Delta v_h \quad (6)$$

Table 2. Typical parameter ranges based on urban form.

Types of Urban Landforms	Average Building Height/m	Zero-Plane Displacement/m
Low-rise sparsely populated suburbs	5–10	0
A district of medium-rise buildings	15–20	0.7H
Central Business district	30–50	0.8H
z_{ref} (a fixed reference level)	10–15	

2.3. Path Movement Model

The dynamic typhoon wind field is used to describe the evolution of the wind field after typhoon landfall, which can be modeled as a series of static wind fields. For the

characterization of the typhoon trajectory, existing studies indicate that the overall path typically exhibits an S-shaped smooth turning pattern, superimposed with small periodic sinusoidal oscillations. Accordingly, the specific trajectory function of the typhoon can be expressed as follows.

$$\begin{cases} L_x = \frac{A_x}{1+\exp(-\eta_1(t-t_0))} \\ L_y = \frac{A_y}{1+\exp(-\eta_2(t-t_0))} \end{cases} \quad (7)$$

$$\begin{cases} \delta x_{\text{osc}} = \alpha_x \cdot \sin(\omega t + \phi) \\ \delta y_{\text{osc}} = \alpha_y \cdot \sin(\omega t + \phi) \end{cases} \quad (8)$$

$$\begin{cases} x(t) = x_0 + L_x + \delta x_{\text{osc}} \\ y(t) = y_0 + L_y + \delta y_{\text{osc}} \end{cases} \quad (9)$$

where Equation (7) represents the Logistic-based main trajectory, and Equation (8) denotes the sinusoidal auxiliary trajectory correction factor. By combining these two components, the mathematical expression of the typhoon trajectory after landfall can be obtained. A_x and A_y are the total displacement amplitudes of the main typhoon track in the longitudinal and latitudinal directions (km), reflecting the overall movement range from landfall to dissipation; η_1 and η_2 are the steepness parameters of the Logistic function (h^{-1}), controlling the rapidity of the turning process near the reference time t_0 (typically taken as the landfall time); α_x and α_y are the amplitudes of the periodic sinusoidal perturbations (km), characterizing the magnitude of short-term track deflections induced by terrain or large-scale circulation; ω is the angular frequency of the perturbation (rad/h); and ϕ is the initial phase (rad).

3. Modeling the Impact of Extreme Weather on Distribution Network Equipment

3.1. Distribution Network Framework Fault Scenario Simulation

During typhoon conditions, the conductors and pylons of urban distribution networks are highly susceptible to typhoon damage. Damage to pylons can cause faults between node circuits, subsequently leading to outages in the load areas served by those circuits.

3.1.1. Probability Model for Distribution Network Fault Occurrence

1. Line component strength

The strength of conductors and poles is subject to uncertainty, typically following a normal distribution. Considering that strength quantities cannot physically take negative values, we adopt a truncated normal distribution on the domain $[0, +\infty)$ for modeling, with the probability density function given by:

$$f_S(S_{l/p}) = \frac{1}{\sqrt{2\pi}\delta_{l/p}1 - \Phi(-\mu_{l/p}/\delta_{l/p})} \exp\left[-\frac{1}{2}\left(\frac{S_{l/p} - \mu_{l/p}}{\delta_{l/p}}\right)^2\right] \quad (10)$$

where $S_{l/p}$ denotes the strength of conductor l or tower p ; $\mu_{l/p}$ denotes the mean strength of conductor l or tower p ; $\delta_{l/p}$ denotes the strength error of conductor l or tower p , and the term $1 - \Phi(-\mu_{l/p}/\delta_{l/p})$ in the denominator is the renormalization factor for the truncated normal distribution.

The mean strength values are determined by the rated mechanical strength of the corresponding conductor or pole/tower type, while the standard deviations describe manufacturing tolerance, aging, corrosion, installation error, and other uncertainties. When

local historical fault records are available, these distribution parameters can be calibrated using observed component-damage samples.

2. Line Component Load Effect

Under typhoon conditions, the loads acting on conductors mainly include horizontal wind loads and gravitational loads. The loads acting on poles and towers primarily consist of the horizontal wind loads transmitted from the suspended conductors and the wind loads acting on the pole or tower body itself. When these loads are applied to conductors and poles/towers, they induce cross-sectional stress in the conductors and bending moments in the pole or tower sections, which are collectively referred to as load effects.

The tension in a conductor is typically maximized at the highest suspension point. Therefore, the combined tension at this point is used in this study to calculate the cross-sectional stress of the conductor, which can be expressed as follows.

$$L_{l,t} = \frac{F_{l,t}}{A} \quad (11)$$

where $L_{l,t}$ denotes the cross-sectional stress of the conductor at time t ; $F_{l,t}$ represents the combined tension at the highest suspension point of the conductor at time t ; and A is the effective cross-sectional area of the conductor. The combined tension $F_{l,t}$ consists of the horizontal wind load component $F_{l1,t}$ and the vertical gravitational load component $F_{l2,t}$, which can be expressed as follows.

$$F_{l,t} = \sqrt{(F_{l1,t})^2 + (F_{l2,t})^2} \quad (12)$$

$$F_{l1,t} = \frac{V_t^2}{1600} \delta_w \delta_{lb} D_l l_w \sin^2 \theta \quad (13)$$

$$F_{l2,t} = m_0 g l_w \quad (14)$$

where δ_w denotes the wind pressure non-uniformity coefficient; δ_{lb} represents the conductor profile coefficient; D_l is the conductor outer diameter; l_w denotes the tower span length; θ is the angle between the wind direction and the conductor orientation ($^\circ$). When $\theta = 90^\circ$, $\sin^2 \theta = 1$, and the wind load reaches its maximum; when $\theta = 0^\circ$ or 180° , the wind load approaches zero. The typical angle between typhoon wind direction and line orientation in coastal areas ranges from 30° to 90° .

The horizontal wind loads on conductors borne by the tower are as follows:

$$F_{p,t} = F_{l1,t} \quad (15)$$

Wind loads on the mast are as follows:

$$T_{p,t} = \frac{V_t^2}{1600} \delta_V \delta_{pb} \delta_h \frac{D_0 + D_p}{2} h_p \quad (16)$$

where δ_V denotes the wind vibration coefficient; δ_{pb} represents the tower shape coefficient; δ_h signifies the wind pressure height variation coefficient; D_0 and D_p denote the pole tip diameter and base diameter respectively; h_p indicates the tower height.

The wind-load coefficients in Equations (13) and (16), including the wind-pressure non-uniformity coefficient, conductor shape coefficient, wind vibration coefficient, tower shape coefficient, and height variation coefficient, are selected according to standard wind-load calculation procedures for overhead distribution lines and pole/tower structures. In applications, these coefficients can be further calibrated using local design codes, conductor type, pole/tower geometry, installation height, and historical typhoon fault records.

3. Probability Model for Line Component Failures

According to structural reliability theory, failure may occur in components when the tensile strength of conductors or the bending strength of towers is less than the stress in the conductor cross-section or the bending moment in the tower cross-section. Therefore, the failure probabilities for conductors and towers are as follows:

$$p_{ij,k_{l/p},t} = P\left\{S_{l/p} < L_{l/p,t}\right\} = \frac{\int_0^{L_{l/p,t}} \frac{1}{\sqrt{2\pi}\delta_{l/p}} \exp\left[-\frac{1}{2}\left(\frac{S_{l/p} - \mu_{l/p}}{\delta_{l/p}}\right)^2\right] dS_{l/p}}{1 - \Phi(-\mu_{l/p}/\delta_{l/p})} \quad (17)$$

where $p_{ij,k_{l/p},t}$ denotes the failure rate of conductor in distribution line or the failure rate of pole in distribution line. For typical engineering ranges with $\delta/\mu < 0.25$, the renormalization factor $1 - \Phi(\delta/\mu)$ is close to unity, and the numerical difference between the truncated and full normal distributions is negligible. The failure threshold follows structural reliability theory: a conductor or pole/tower is regarded as failed when the calculated load effect exceeds the corresponding mechanical strength.

3.1.2. Failure Simulation Algorithm Considering Pole–Tower Correlation in Distribution Network Structures

When a transmission line is damaged or a pole/tower collapses at a certain location during a typhoon event, the adjacent lines or poles/towers are often affected by the initial failure point, resulting in a significant increase in their failure probabilities. The distribution of such failure probabilities can be simulated in a manner similar to the Independent Cascade Model commonly used in social network diffusion studies. The flowchart illustrating the specific algorithm simulation is shown in Figure 1.

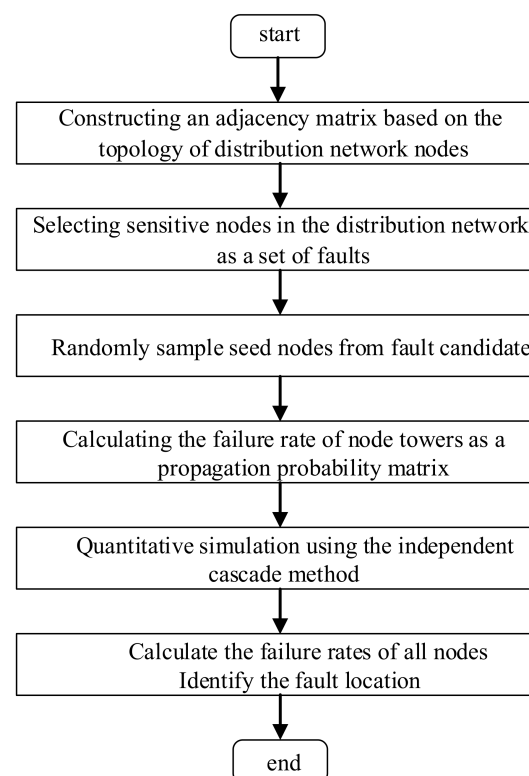


Figure 1. Flowchart for fault location on power distribution lines.

In urban distribution networks, due to the large number of poles/towers and the relatively short distances between adjacent distribution lines, the poles/towers can be approximately represented as discretely distributed and serially connected components near network nodes. The failure probability of multiple poles/towers connected in series is then used as an element of the failure probability propagation matrix. Assuming that there are n poles/towers distributed around a node v , the corresponding failure propagation probability can be expressed as follows.

$$p_{an,v} = 1 - \prod_{k_p=1}^n (1 - p_{ij,k_p,t}) \quad (18)$$

Suppose at time t , node v has n neighboring nodes activated at time $t - 1$: $a_1, a_2, \dots, a_n$. The probabilities that these nodes activate node v are respectively $p_{a1,v}, p_{a2,v}, \dots, p_{an,v}$. Then the probability that node v is activated is as follows:

$$P_v = 1 - (1 - p_{a1,v})(1 - p_{a2,v}) \cdots (1 - p_{an,v}) \quad (19)$$

3.1.3. Simulation of Distribution Network Fault Scenarios Based on Copula Joint Modeling

Within the Copula-based joint modeling framework, the construction of marginal random variables is a prerequisite for achieving multi-source information fusion. According to Sklar's theorem, joint distribution can be constructed without altering the marginal distribution forms of the individual random variables. Therefore, this study first performs marginal modeling for the pole/tower failure frequency and the conductor failure probability, respectively, and subsequently normalizes them into the domain required by the Copula function.

To characterize the failure behavior of poles and towers, an additive aggregation approach is adopted to construct an equivalent line-level pole/tower failure frequency index, which is expressed as follows.

$$\Lambda_{ij}^t = \sum_{k \in \mathcal{T}_{ij}} \lambda_k \quad (20)$$

where $\mathcal{T}_{ij}$ denotes the set of poles/towers associated with node; λ_k represents the pole/tower failure frequency, which is quantitatively simulated by the Independent Cascade algorithm under a given disaster scenario.

Since the analytical distribution form of Λ_{ij}^t is generally difficult to assume a priori, a nonparametric statistical method is adopted in this study to estimate its marginal distribution. Specifically, under a given set of disaster scenarios or multiple simulation samples, the empirical distribution function of Λ_{ij}^t is constructed and subsequently mapped into a random variable defined on the unit interval $[0, 1]$ through the probability integral transform.

$$F_{\Lambda}(x) = \frac{1}{N} \sum_{n=1}^N \mathbb{I}(\Lambda_{ij,n}^t \leq x) \quad (21)$$

where N denotes the sample size; $\Lambda_{ij,n}^t$ represents the equivalent tower failure frequency under the n th disaster scenario, with the random variable $F_{\Lambda}(x)$ denoted as U_{ij} .

For conductors, their failure behavior during a single disaster event is treated as a random occurrence. Let the probability of conductor failure be denoted as p_{ij}^L , employing a Bernoulli distribution $V_{ij} \sim \text{Bernoulli}(p_{ij}^{\text{line}})$ to describe the single binary random event.

The Gumbel copula belongs to the Archimedean family and excels at capturing upper-tail dependence—namely, the characteristic that when the pole failure frequency is high (severe pole damage under extreme wind speeds), the probability of conductor failure

risks synchronously. Compared with the Clayton (lower-tail dependence) and Gaussian (no tail dependence) copulas, the Gumbel better fits the physical mechanism of “strong wind simultaneously damaging both poles and conductors” during typhoon events. This selection has been confirmed via the Akaike Information Criterion (AIC) after comparing the three candidate copulas. It is now explicitly stated that $\theta \in [1, \infty)$, with $\theta = 1$ indicating independence and $\theta \rightarrow \infty$ indicating perfect dependence; the relationship with Kendall’s τ , $\tau = 1 - 1/\theta$, is also clarified.

Given the correlation of failure events within high-risk intervals during disaster scenarios, the Gumbel copula is employed to model joint failure events.

$$C_\theta(u, v) = \exp\left(-\left[(-\ln u)^\theta + (-\ln v)^\theta\right]^{1/\theta}\right), \theta \in [1, \infty) \quad (22)$$

where the association parameter θ is related to Kendall’s rank correlation coefficient τ via $\tau = 1 - 1/\theta$. Based on historical or simulated samples of pole failure frequency and conductor failure status, the sample estimate is computed, and a moment estimate of θ is obtained.

Copula sampling with mixed marginals and fault scenario generation: Since U_{ij} (empirical CDF values of pole failure frequency) is continuous while V_{ij} (conductor failure status) is Bernoulli-type, their mixed marginals require special treatment within the standard copula framework. We adopt the jittering continuous extension approach.

Step 1: Generate s sets of coupled uniform random variables (u_s, v_s) , $s = 1, 2, \dots, S$.

Step 2: For each sample pair, map u_s to the pole failure frequency F_s via the inverse empirical CDF of pole failure frequency; compare v_s with a pre-defined threshold (e.g., the quantile corresponding to the marginal probability of conductor failure p_f) to determine the conductor failure status $B_s \in (0, 1)$ (0: normal; 1: failure) in that scenario.

Step 3: All S joint samples (F_s, B_s) are taken as input and successively fed into the subsequent distribution network fault impact analysis model, generating the corresponding set of faulted lines to form the complete fault scenario ensemble.

3.2. Modeling the Contribution of Distributed Resources Under Extreme Weather Conditions

3.2.1. Modeling Wind Turbine Output Under Extreme Weather Conditions

A wind turbine output model is established using the constructed typhoon wind field model. The output characteristics of wind turbines are typically represented by three parameters: cut-in wind speed v_{in} , rated wind speed v_r , and cut-out wind speed v_{out} . Actual wind speeds at wind farms may vary slightly due to geographical differences. To simplify the output model, it is assumed that all wind turbines within the same wind farm experience identical wind direction and wind speed. The wind farm is thus treated as a single equivalent wind turbine, whose output model can be expressed as:

$$P_{wind} = \begin{cases} 0 & 0 \leq v < v_{in} \\ (A + Bv + Cv^2)P_r & v_{in} \leq v < v_r \\ P_r & v_r \leq v < v_{out} \\ 0 & v \geq v_{out} \end{cases} \quad (23)$$

where P_{wind} denotes the output power of the wind turbine at wind speed v ; P_r represents the rated power of the wind turbine; v indicates the current wind speed. The specific values of the constants A , B and C are given with reference to the parameters of a typical doubly fed wind turbine.

3.2.2. Modeling Photovoltaic Power Output Under Extreme Weather Conditions

Under typhoon conditions, photovoltaic output is significantly affected by extreme meteorological factors such as strong winds, heavy rainfall, and rapid cloud cover changes, resulting in substantial deviations between actual output and forecast values. To accurately characterize photovoltaic output uncertainty during such extreme weather, this paper introduces a box-type uncertainty set U to describe its fluctuation range, based on the predicted photovoltaic output P_i^{PV*} . The specific expression is as follows:

$$U = \left\{ (1 - \tau)P_i^{PV*} \leq P_i^{PV} \leq P_i^{PV*} \right\} \quad (24)$$

where P_i^{PV*} denotes the forecast value for photovoltaic output; τ represents the uncertainty parameter, used to adjust the upper and lower bounds of the uncertainty set. During typhoon conditions, actual photovoltaic output typically falls below the forecast value due to sudden drops in irradiance and potential shading or damage to modules. Consequently, the lower bound of the uncertainty set is set to $(1 - \tau)P_i^{PV*}$, reflecting the risk of significantly reduced output during extreme weather events. The parameter τ characterizes the uncertainty radius of PV output. Under typhoon conditions, actual output is typically lower than the forecast value; statistics show that sudden irradiance reduction can cause output drops of 30–70%. In the base case, $\tau = 0.5$ is adopted.

4. Multi-Dimensional Collaborative Scheduling Recovery Strategy

In the pre-disaster prevention phase, this study pre-plans the deployment locations and quantities of mobile energy storage systems. This accounts for extreme photovoltaic output scenarios and the potential for network reconfiguration, operating under conditions where both the timing and duration of disasters remain uncertain. The aim is to ensure rapid emergency response deployment post-disaster, supporting the load restoration process. Wind turbines are also considered to be recovery resources. Upon transitioning to the post-disaster recovery phase, a multi-source coordinated restoration optimization model is established. This model centrally coordinates the dispatch of distributed resources such as mobile energy storage and electric vehicles, achieving optimized power allocation across both temporal and spatial dimensions. Consequently, critical load supply is restored to the greatest extent possible, thereby enhancing the resilience of the distribution network. The flowchart for the specific modeling and solution process is shown in Figure 2.

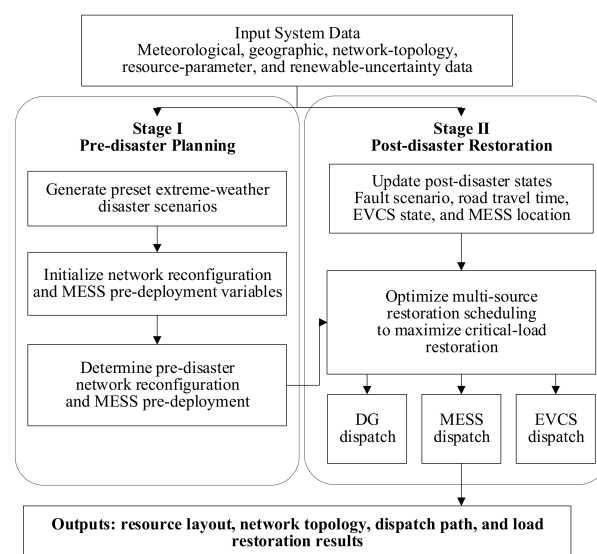


Figure 2. Two-stage pre-disaster/post-disaster optimization framework.

4.1. Pre-Disaster Preparedness Layout Model

4.1.1. Objective Function

In pre-disaster planning, an optimization model is established to determine the configuration quantity and location of mobile energy storage, with the objective of minimizing both the configuration costs and load curtailment costs under the most adverse distributed power generation output scenario.

$$\min_{\alpha_{i,0}^{\text{ME}}, \alpha_{ij}} \max_{P_i^{\text{PV}}} \min_{P_i^{\text{Lshd}}, Q_i^{\text{Lshd}}, P_i^{\text{DG}}, Q_i^{\text{DG}}, P_{ij}, Q_{ij}} \sum_{i \in \Omega_N} (C^{\text{ME}} \alpha_{i,0}^{\text{ME}} + w_i P_i^{\text{Lshd}}) \quad (25)$$

where $\alpha_{i,0}^{\text{ME}}$ denotes a Boolean variable, with a value of 0 indicating that the mobile energy storage unit is not connected to the current node i , and a value of 1 indicating that the mobile energy storage unit is connected to the current node i , with precisely one such unit connected to node i ; α_{ij} denotes a Boolean variable representing the open/closed state of a distribution line, where 0 indicates the line is open and 1 indicates the line is closed. This variable reflects the reconfiguration state of the distribution network topology; Ω_N denotes the set of nodes in the distribution network; P_i^{PV} denotes photovoltaic output; P_i^{Lshd} denotes active power load shedding, a decision variable; Q_i^{Lshd} denotes reactive power load shedding, a decision variable; P_{ij} denotes active power on the distribution line; Q_{ij} denotes reactive power on the distribution line; P_i^{DG} denotes the active power output from wind turbines connected to distribution network node i ; Q_i^{DG} denotes the reactive power output from wind turbines connected to distribution network node i ; C^{ME} denotes the configuration cost of a single mobile energy storage unit during pre-disaster deployment, encompassing installation and transportation expenses; and w_i represents the unit load reduction cost at distribution network node i .

4.1.2. Constraints

1. Output Constraints for Distributed Generation

The output capacity of all distributed power generation nodes is constrained by their rated capacity. For mobile energy storage units and electric vehicles operating at full charge prior to the disaster, this upper limit is uniformly defined as their rated charge/discharge power. Concurrently, the model incorporates operational constraints for active power, reactive power, and power factor, with photovoltaic and wind turbine systems operating under fixed power factor modes. Specific constraints are as follows:

$$-\sum_{i \in \{M, E, P, T\}} S_{i, \max}^{\text{DG}} \leq S_i^{\text{DG}} \leq \sum_{i \in \{M, E, P, T\}} S_{i, \max}^{\text{DG}} \quad (26)$$

$$\cos \varphi_{i, \min}^{\text{DG}} \leq \frac{P_i^{\text{DG}}}{\sqrt{(P_i^{\text{DG}})^2 + (Q_i^{\text{DG}})^2}} \leq \cos \varphi_{i, \max}^{\text{DG}} \quad (27)$$

where $S_{i, \max}^{\text{DG}}$ denotes the maximum power output of distributed resources; $\cos \varphi_{i, \min}^{\text{DG}}$, $\cos \varphi_{i, \max}^{\text{DG}}$ denotes the minimum power factor of distributed resources; M , E , P , and T denote the node sets for mobile energy storage, electric vehicle charging stations, photovoltaic systems, and wind turbines respectively.

2. Output Constraints for Mobile Energy Storage

The pre-deployment of mobile energy storage is constrained by two factors: firstly, the total volume of resources pre-positioned prior to disasters is limited, and all equipment shares identical operational parameters; secondly, the maximum number of devices pre-

configured at each node is capped at one unit. The specific constraint expressions are as follows:

$$\sum_{i \in \Omega_N} \alpha_{i,0}^{\text{ME}} \leq N_{\text{max}}^{\text{ME}} \quad (28)$$

where $N_{\text{max}}^{\text{ME}}$ denotes the maximum number of mobile energy storage vehicles deployed during the pre-disaster planning phase of the distribution network.

3. Radiation Topology Constraints in Distribution Networks

To address the integration of potential isolated islands (including those without power supply) with the main grid during restoration, the proposed model employs an enhanced single-commodity flow method to ensure radial system operation. This framework achieves optimized control over the number of isolated islands by introducing virtual source nodes and associated 0–1 variables, ultimately applying the Big-M method to linearize and relax the resulting virtual power flow constraints. The specific constraint expressions are as follows:

$$\sum_{ij \in \Omega_B} \alpha_{ij} = N_B - \sum_{i \in \Omega_N} S_i^{\text{VS}} \quad (29)$$

$$\sum_{j \in \delta(i)} F_{ij} - \sum_{k \in \gamma(i)} F_{ki} = F_i^{\text{VS}} - 1 \quad (30)$$

$$-M_1 S_i^{\text{VS}} \leq F_i^{\text{VS}} \leq M_1 S_i^{\text{VS}} \quad (31)$$

$$-M_1 \alpha_{ij} \leq F_{ij} \leq M_1 \alpha_{ij} \quad (32)$$

where Ω_B denotes the set of all distribution lines in the distribution network; N_B denotes the set of all nodes in the distribution network; $\delta(i)$ denotes the set of child nodes of a node in the distribution network; $\gamma(i)$ denotes the set of parent nodes of a node in the distribution network; F_{ij} denotes the virtual power flow of a distribution line in the distribution network; F_i^{VS} denotes the virtual power flow emitted by a node in the distribution network; and M_1 denotes the maximum number in the Big-M method.

4. Distribution Network Operational Constraints

The operational constraint set constructed for the distribution network centers on the DistFlow power flow equations to characterize its radial operational characteristics. To address topology reconfiguration issues arising from changes in line switch states, the model employs the Big-M method to effectively relax the nonlinear voltage equations. The specific constraint expressions are as follows:

$$\begin{cases} \sum_{j \in \delta(i)} P_{ij} - \sum_{j \in \gamma(i)} (P_{ji} - R_{ji} I_{ji}^{\text{sq}r}) = P_i^{\text{DG}} - P_{i,\text{max}}^{\text{L}} + P_i^{\text{Lshd}} \\ \sum_{j \in \delta(i)} Q_{ij} - \sum_{j \in \gamma(i)} (Q_{ji} - X_{ji} I_{ji}^{\text{sq}r}) = Q_i^{\text{DG}} - Q_{i,\text{max}}^{\text{L}} + Q_i^{\text{Lshd}} \end{cases} \quad (33)$$

$$\begin{aligned} V_i^2 - V_j^2 - 2(R_{ij}P_{ij} + X_{ij}Q_{ij}) + \\ (R_{ij}^2 + X_{ij}^2) \frac{P_{ij}^2 + Q_{ij}^2}{V_i^2} \leq M_1(1 - \alpha_{ij}) \end{aligned} \quad (34)$$

$$\begin{aligned} V_i^2 - V_j^2 - 2(R_{ij}P_{ij} + X_{ij}Q_{ij}) + \\ (R_{ij}^2 + X_{ij}^2) \frac{P_{ij}^2 + Q_{ij}^2}{V_i^2} \geq M_1(\alpha_{ij} - 1) \end{aligned} \quad (35)$$

$$V_{i,\text{min}} \leq V_i \leq V_{i,\text{max}} \quad (36)$$

$$0 \leq I_{ij}^2 \leq I_{ij,\text{max}}^2 \quad (37)$$

$$P_{ij}^2 + Q_{ij}^2 \leq V_i^2 I_{ij}^2 \quad (38)$$

5. Load reduction constraints at distribution network nodes

Assuming the power factor of the load at the distribution network node is a fixed value, the specific expression is as follows:

$$0 \leq P_i^{\text{Lshd}} \leq P_{i,\max}^{\text{L}} \quad (39)$$

$$Q_i^{\text{Lshd}} = \frac{Q_{i,\max}^{\text{L}}}{P_{i,\max}^{\text{Lshd}}} P_i^{\text{Lshd}} \quad (40)$$

where $P_{i,\max}^{\text{L}}$ denotes the maximum active power load at distribution network nodes; $Q_{i,\max}^{\text{L}}$ denotes the maximum reactive power load at distribution network nodes.

By solving this model, one can derive the optimal configuration and network topology for mobile energy storage under extreme photovoltaic scenarios. The resulting system layout will ensure that mobile energy storage can be deployed immediately post-disaster to maximize the restoration of critical loads.

4.2. Post-Disaster Recovery Model

4.2.1. Objective Function

Following a disaster, the distribution network will disconnect from the main grid and enter an islanded operation mode. At this stage, the duration of the fault can be estimated based on the severity of the disaster and the availability of repair resources. Accordingly, a multi-source collaborative restoration model is established with the objective of minimizing the weighted load reduction during this period. By dynamically dispatching distributed resources such as mobile energy storage, electric vehicles, and diesel generators, the model aims to maximize the resilience of the distribution network. The specific objective function is as follows:

$$\min \sum_{t \in \Omega_T} \sum_{i \in \Omega_N} w_i P_{i,t}^{\text{Lshd}} \quad (41)$$

where $P_{i,t}^{\text{Lshd}}$ denotes the active power load shedding at distribution network node i at time t ; Ω_T represents the set of time intervals following the fault occurrence. Considering the temporal scale of distribution network dispatch and the spatiotemporal nature of mobile energy storage under extreme weather conditions, the dispatch interval is typically set to one hour.

During multi-source restoration following disasters, the connection status of mobile energy storage to distribution network nodes, the output of distributed resources, the output of electric vehicle charging stations, and load reduction power are treated as decision variables to be determined for each time interval.

4.2.2. Constraints

1. Mobile Energy Storage Dispatch and Output Constraints

The scheduling of mobile energy storage exhibits typical spatio-temporal coupling characteristics within the context of grid-transport network integration, wherein its state is jointly determined by charging/discharging on the power side and mobility behavior within the transport network. Specifically, the model must account for transit times between nodes and installation configuration times, assuming no energy consumption during transportation. The reliability of mobile energy storage scheduling during disasters hinges critically on addressing the time-varying uncertainties of the transport network. Given that typhoon disasters systematically alter road conditions, causing variations in

transit times, this paper constructs a transport integration coefficient to precisely reflect the dynamic spatial relationship between actual vehicle speeds and equivalent travel distances. The constraints governing the transit time $T_{i,j,k}^{ME}(t)$ between nodes i and j , the equivalent transport distance $D_{j,k}(t)$, and the actual vehicle speed $v_i^{ME}(t)$ for mobile energy storage are as follows:

$$T_{i,j,k}^{ME}(t) = \frac{D_{j,k}(t)}{v_i^{ME}(t)} \quad (42)$$

$$D_{j,k}(t) = D_{j,k,0}(1 + \gamma \cdot c(t)) \quad (43)$$

$$v_i^{ME}(t) = v_{i,0}e^{-1.7c} \quad (44)$$

where $D_{j,k,0}$ denotes the distance between distribution network nodes; c represents the level of traffic congestion during typhoon scenarios, which correlates with the severity of disaster impacts and urban travel planning demands, and can be estimated post-disaster based on the extent of damage to the transport network and historical experience; $v_{i,0}$ indicates the ideal traffic flow velocity under optimal conditions. When $c(t) \in [0,1]$ is the traffic congestion index (0: free flow, 1: fully congested); $c_r(t) \in \{0,1\}$ is a road disruption indicator, taking the value 1 only when a road segment is disrupted and requires rerouting; γ is the detour distance amplification factor. Under normal congestion scenarios, $c_r(t) = 0$, and the travel time is affected solely by speed reduction, with no double-counting.

Mobile energy storage may connect to no more than one node at any given time, with the specific constraint expressed as follows:

$$\sum_{j \in \Omega_N} \alpha_{i,j,l}^{ME} \leq 1 \quad (45)$$

Moreover, mobile energy storage may only contribute power output when it remains connected throughout an entire dispatch period. The specific constraint expression is as follows:

$$\alpha_{i,j,t}^{ME} + \alpha_{i,k,t+\Delta t}^{ME} \leq 1 \quad \forall \Delta t < T_{i,j,k}^{ME}(t) + T_0^{ME} \quad (46)$$

The state of charge and discharge of mobile energy storage systems is determined by their own state of charge, with the specific constraints expressed as follows:

$$E_{i,t+\Delta t}^{ME} = E_{i,t}^{ME} + P_{i,t}^{Mch} \eta_i^{Mch} \Delta t - \frac{P_{i,t}^{Mdch}}{\eta_i^{Mdch}} \Delta t \quad (47)$$

$$E_{i,min}^{ME} \leq E_{i,t}^{ME} \leq E_{i,max}^{ME} \quad (48)$$

where $E_{i,max}^{ME}$ denotes the upper limit of mobile energy storage capacity; $E_{i,min}^{ME}$ denotes the lower limit of mobile energy storage capacity; η_i^{Mch} denotes the charging efficiency of mobile energy storage; η_i^{Mdch} denotes the discharging efficiency of mobile energy storage; $P_{i,t}^{Mch}$ denotes the active power during mobile energy storage charging; and $P_{i,t}^{Mdch}$ denotes the active power during mobile energy storage discharging.

Furthermore, the charging and discharging power of the mobile energy storage is subject to certain constraints, expressed as follows:

$$\begin{cases} 0 \leq P_{i,t}^{Mch} \leq U_{i,t}^{Mch} P_{i,max}^{Mch} \\ 0 \leq P_{i,t}^{Mdch} \leq U_{i,t}^{Mdch} P_{i,max}^{Mdch} \\ U_{i,t}^{Mch} + U_{i,t}^{Mdch} \leq 1 \end{cases} \quad (49)$$

3. Power Output Constraints for Electric Vehicle Charging Stations

Assuming both electric vehicle charging points and electric vehicles satisfy the conditions for vehicle–grid interaction, electricity can be fed back into the grid. Following a disaster, electric vehicles connected to charging points can discharge their batteries to supply power to loads. Furthermore, to systematically coordinate post-disaster load restoration with the personalized mobility needs of electric vehicle users, this model treats electric vehicle fleets as a class of mobile energy storage resources possessing both spatiotemporal coupling characteristics and social behavioral attributes. Traditional dispatch models solely consider their physical charging and discharging constraints. This paper, however, introduces the concept of the ‘mobility chain’ to construct a more refined dispatchable capacity assessment framework. The specific constraint expressions are as follows:

$$\begin{cases} 0 \leq P_{i,t}^{Ech} \leq U_{i,t}^{Ech} P_{i,max}^{Ech} \\ 0 \leq P_{i,t}^{Edch} \leq U_{i,t}^{Edch} P_{i,max}^{Edch} \end{cases} \quad (50)$$

$$U_{i,t}^{Ech} + U_{i,t}^{Edch} \leq 1 \quad (51)$$

$$E_{i,t+\Delta t}^E = E_{i,t}^E + P_{i,t}^{Ech} \eta_i^{Ech} \Delta t - \frac{P_{i,t}^{Edch}}{\eta_i^{Edch}} \Delta t \quad (52)$$

$$E_{i,min}^E(t) = E_{i,tech}^{min} + \sum_{v \in \Phi_i(t)} (D_v^{tr} \cdot \gamma_v^E) \quad (53)$$

where $P_{i,t}^{Ech}$ denotes the charging power of an electric vehicle charging station; $P_{i,t}^{Edch}$ denotes the discharging power of an electric vehicle charging station; $P_{i,max}^{Ech}$ denotes the maximum charging power of the electric vehicle charging station; $P_{i,max}^{Edch}$ denotes the maximum discharging power of the electric vehicle charging station; $U_{i,t}^{Ech}$ is a Boolean variable, where a value of 1 indicates the electric vehicle charging station is in charging mode, and a value of 0 indicates the electric vehicle charging station is not transmitting power; $U_{i,t}^{Edch}$ is a Boolean variable, where a value of 1 indicates the EV charging station is discharging, and a value of 0 indicates no power transfer is occurring; η_i^{Ech} denotes the charging efficiency of the EV charging station; η_i^{Edch} denotes the discharging efficiency of the EV charging station; $\Phi_i(t)$ denotes the set of electric vehicles connected to the charging station with planned journeys after time t ; D_v^{tr} represents the remaining journey distance for vehicle v ; and γ_v^E indicates the specific energy consumption per unit distance for vehicle v . $E_{i,tech}^{min}$ denotes the fixed technical capacity lower bound based on electric vehicle battery lifespan considerations.

4.3. Model Solution

The pre-disaster model (Equation (25)) is a tri-level robust optimization problem with a min–max–min structure, where PV output uncertainty is described by the box-type set (Equation (24)). We adopt the Column-and-Constraint Generation (C & CG) algorithm for its solution. The original problem is decomposed into a Master Problem (MP) and a Subproblem (SP).

The branch power flow constraints in both the pre-disaster and post-disaster models contain quadratic nonlinear terms. Introducing an auxiliary variable $lij = (P_{ij}^2 + Q_{ij}^2)/V_i^2$ representing the squared branch current magnitude, Equation (38) is transformed into the following rotated second-order cone form:

$$\left\| [2P_{ij}, 2Q_{ij}, l_{ij} - V_i^2]^\top \right\|_2 \leq l_{ij} + V_i^2$$

The exactness of this relaxation for radial distribution network optimization has been widely proven, and we verify the relaxation error as a posteriori metric.

The post-disaster restoration model is a single-step or multi-step MISOCP, with integer variables related to MESS/EV dispatch decisions, and can be solved directly. All optimization models are formulated using the YALMIP toolbox in the MATLAB R2024b environment and solved via Gurobi 9.5.2. The MIP gap tolerance is set to 1×10^{-4} .

Under specified grid-transportation network integration system parameters, optimal load shedding and power generation schemes are obtained by solving both the pre-disaster deployment optimization model and the multi-source collaborative post-disaster recovery optimization model. Solution computation employs MATLAB's YALMIP toolbox to invoke the commercial solver Gurobi 8.1.1, yielding highly satisfactory results.

5. Case Study Analysis

5.1. Test System Description

This paper selects a 51-node system in a city along China's southeastern coast as the test system, covering a geographical area spanning longitudes (118.8° E to 119.05° E) and latitudes (25.81° N to 26.1° N). It is assumed that the transport network topology aligns with the power grid topology, and that pre-disaster multi-source distributed resources have sufficient time for pre-deployment. The distribution network optimization dispatch model established herein employs the Gurobi solver within the YALMIP toolbox of MATLAB 2024b for solution.

5.2. Parameter Settings

5.2.1. Simulation Analysis of Distribution Line Faults

This section employs a spatio-temporal typhoon evolution model to simulate typhoon landfall and calculate distribution line fault rates. It approximates pylons near nodes while utilizing an independent cascading algorithm to simulate fault line selection.

The track and intensity parameters of the simulated typhoon event are calibrated against the actual records of Typhoon Meranti (2016) at its landfall in Fujian Province: the central maximum wind speed at landfall is set at 48 m/s, the radius of maximum wind at 40 km, the translation direction at north-northwest, and the translation speed at approximately 20 km/h. The wind field decay parameters are selected based on the Kaplan & DeMaria inland decay model. Although we cannot obtain per-line failure measurement data corresponding one-to-one to this event (for reasons discussed in this section), the above parameter settings ensure that the fault scenarios are statistically consistent with the impact characteristics of a typical strong typhoon in this region.

A spherical coordinate system is established based on the urban distribution network testing system. The center point coordinates at typhoon landfall are set to (119.3° E, 25.2° N), with a latitude and longitude change rate of 0.5. The maximum wind speed radius of the typhoon's wind circle is capped at 40 km, and the maximum expandable radius of the wind circle is set to 100 km. As shown in Figure 3, to clearly illustrate the movement of a typhoon's wind field over a coastal city, we have used blue concentric circles to represent the typhoon's wind field and black curves to represent its path.

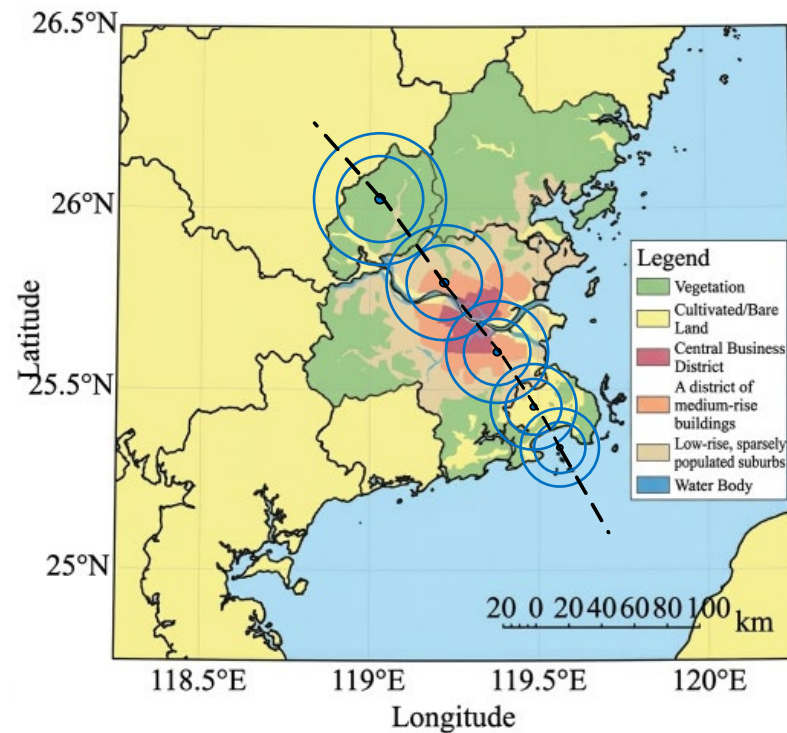


Figure 3. Simulation of a typhoon's landfall.

5.2.2. Distribution Network Fault Simulation

Figure 4 illustrates the distribution of faults on conductors and towers within the distribution network during the 12-period timeframe.

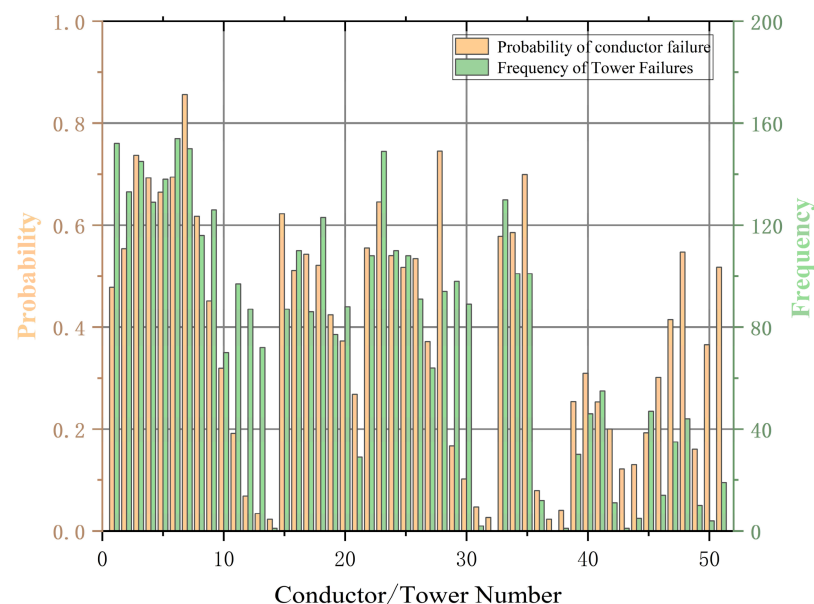


Figure 4. Quantitative diagram of fault probabilities in distribution network structures.

To reflect the stochasticity of typhoons and failures, fault scenarios are independently generated 100 times via the Monte Carlo method. In each simulation, random samples are drawn from the Copula model and the independent cascade failure propagation is executed. Results reported in Sections 5.3 and 5.4 are presented as the mean $\pm$ one standard deviation over the 100 simulations.

The analysis based on Figures 3 and 4 is summarized as follows:

- Taking Line 2 and Line 14 as examples, although the two lines are located relatively close to each other in the topology of the urban distribution network, their failure rates differ significantly when the angle between the wind direction and the conductor orientation is taken into account.
- Lines 26 and 35 are located at the same distance from the center of the typhoon wind circle. Therefore, under an ideal static typhoon wind field model, the wind speeds acting on the two lines would theoretically be identical. However, when considering the ground roughness conditions of the distribution lines and whether the typhoon path passes through urban building clusters, the actual wind speeds experienced by the two distribution lines differ, resulting in different time-varying failure rates.
- Considering Tower 23 and 24, although the two structures are located in close proximity and thus experience approximately the same wind speed, the interdependence between poles/towers leads to different failure characteristics. Since Pole/Tower 23 is connected to a larger number of branches, more adjacent poles/towers are affected when a failure occurs, resulting in a significantly higher failure frequency.

From the simulation results of the distribution network infrastructure failure scenarios, it can be observed that the topological position of the network structure, geographical environment, typhoon wind direction, and intrinsic line parameters all constitute important influencing factors of the distribution network failure scenarios. The advantages of employing the Independent Cascade algorithm to simulate the correlation among poles/towers between nodes and the Copula-based joint modeling approach to characterize the dependency between poles/towers and lines will be further demonstrated in the pre-disaster resource pre-allocation analysis presented in the next section.

5.3. Pre-Disaster Preparedness Analysis

The deterministic optimization approach was employed in the pre-disaster layout plan formulation. The specific layout outcomes and distribution network restructuring strategy are illustrated in Figure 5 below.

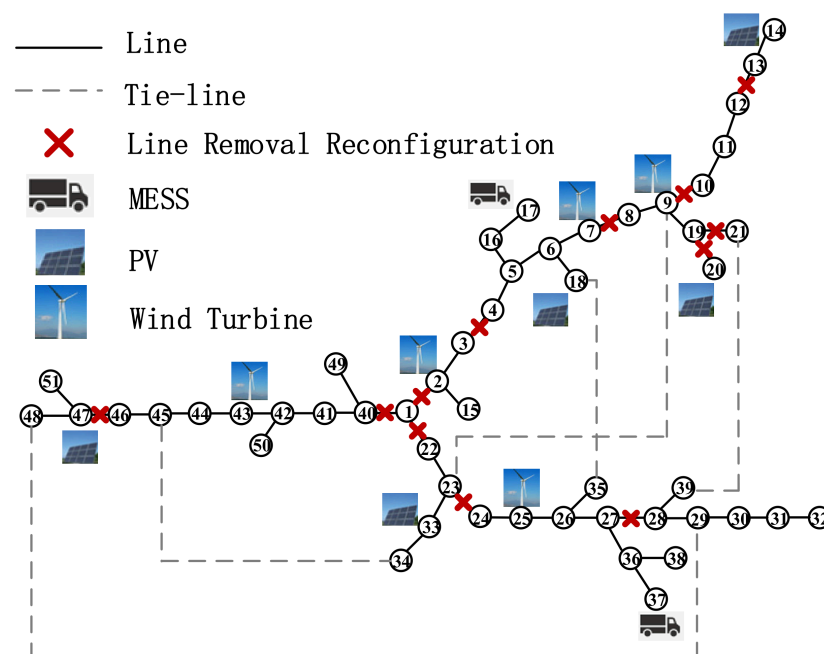


Figure 5. Diagram of proactive restructuring of the distribution network and pre-deployment of resources prior to a disaster.

Since the two mobile energy storage systems (MESSs) are both located in the largest islanded area, this island can obtain higher power supply capability and energy regulation capacity at the early stage of the disaster, making it the “primary restoration island.” Subsequently, as transportation conditions improve or damaged lines are gradually repaired, the MESS units become capable of providing cross-island support to other islands. Therefore, the pre-disaster allocation strategy essentially establishes a stage-wise energy restoration pathway for the post-disaster recovery process: initially concentrating resources on the largest island, followed by gradually expanding restoration support to other islands. This forms a multi-stage restoration mechanism characterized by a “primary–secondary–system-wide” recovery sequence.

To evaluate the effectiveness of the proposed pre-disaster allocation strategy, two scenarios are designed. Scenario 1 considers the pre-allocation of mobile energy storage, whereas Scenario 2 does not consider the pre-allocation of mobile energy storage. The corresponding results are presented in Table 3.

Table 3. Comparison of pre-disaster planning schemes: layout position and costs.

Scenario	Layout Position	Faulty Line	Pre-Layout Costs
1	1, 1	1, 3, 7, 28, 35	29,948.6
2	17, 37	1, 3, 7, 28, 35	23,397.35

Note: Numbers denote the node IDs where MESS units are positioned. E.g., ‘17, 37’ means two MESS units are connected to Nodes 17 and 37, respectively; ‘1, 1’ means both units are connected to the root Node 1.

In terms of pre-allocation cost, the cost of Scenario 2 is reduced by 21.87% compared with Scenario 1. In Scenario 2, the mobile energy storage units are pre-allocated to the root node (Node 1), which reduces the deployment cost of mobile energy storage. However, during the post-disaster restoration stage, the dispatching cost of mobile energy storage increases accordingly. As shown in Table 3, the load shedding cost also increases, indicating that the pre-disaster allocation strategy can effectively reduce the risk-related cost faced by the distribution system.

To further verify the effectiveness of the Independent Cascade algorithm considering correlation effects and the Copula-based joint modeling approach, four groups of comparative experiments are designed. Among them, Scenario 1 corresponds to the proposed model in this paper, where correlation effects are fully considered. Specifically, the Independent Cascade algorithm is used to simulate the correlation of poles/towers between nodes, while the Copula-based joint modeling method is applied to characterize the dependency between conductors and poles/towers. Scenario 2 considers partial correlation effects and only models the inter-node pole/tower correlation. Scenario 3 also considers partial correlation effects but only models the dependency between conductors and poles/towers. Scenario 4 does not consider any correlation effects and directly uses the time-varying distribution network failure scenarios derived from the typhoon simulation. The results are presented in Table 4.

Compared with Scenarios 1, 2, and 3, Scenario 4 directly treats the high-risk lines obtained from the typhoon simulation as the subsequent failure scenarios. In doing so, it neglects the structural correlation of the distribution network, namely the indirect influence of critical lines that affect the overall network operation and the cascading impact of directly high-risk lines under disaster conditions on other lines. As a result, the number of potential islands in the failure scenarios is relatively small, leading to an increase in node load shedding costs. Compared with Scenario 1, both Scenario 2 and Scenario 3 consider correlation effects only partially. Consequently, they fail to accurately identify the truly high-risk lines that play a critical role in distribution network operation, which similarly results in an increase in the cost of pre-disaster resource allocation.

Table 4. Comparison of the effectiveness of pre-disaster planning schemes.

Scenario	Faulty Line	Pre-Layout Costs	Load Reduction Costs
1	1, 3, 7, 28, 35	23,397.35	21,897.35
2	1, 3, 35, 48	25,174.15	23,674.15
3	1, 7, 28, 36	24,147.0	22,647.0
4	3, 28, 35	26,113.25	24,613.25

Note 1: The difference between “Pre-Layout Costs” and “Load Reduction Costs” in the table is fixed at 1500, corresponding to the fixed deployment cost of two MESS units (750 per unit), as all four scenarios involve the same number (two) of MESS units. Note 2: Bold values indicate the scenario with the lowest load reduction cost. This scenario simultaneously considers inter-pole cascading effects (Independent Cascade) and conductor-pole correlation effects (Copula), thereby more accurately identifying high-risk lines and reducing subsequent load curtailment.

5.4. Analysis of Multi-Source Recovery Outcomes Following the Disaster

Following a disaster, the coordinated operation of diverse distributed resources within the urban distribution network.

Based on an analysis of Figures 6–8 below, when the disaster occurs, it is assumed that electric vehicle charging stations (EVCSs) are planned at nodes 5, 15, and 21, while mobile energy storage systems (MESSs) are pre-allocated at nodes 17 and 37. Under this configuration, the entire distribution network is divided into five islands.

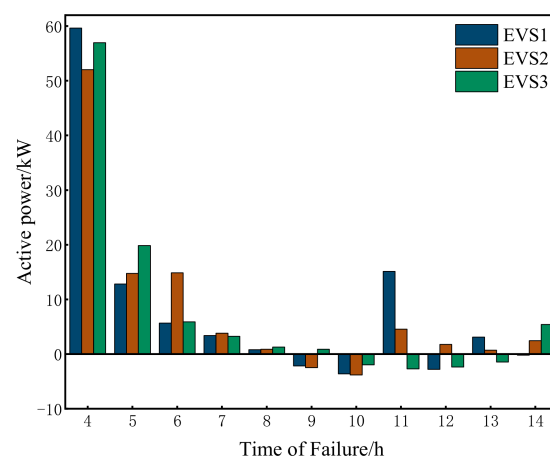


Figure 6. Power transmission of EV charging stations by time period. Positive values indicate EVCS discharging to the grid (V2G mode); negative values indicate EVCS charging from the grid (G2V mode).

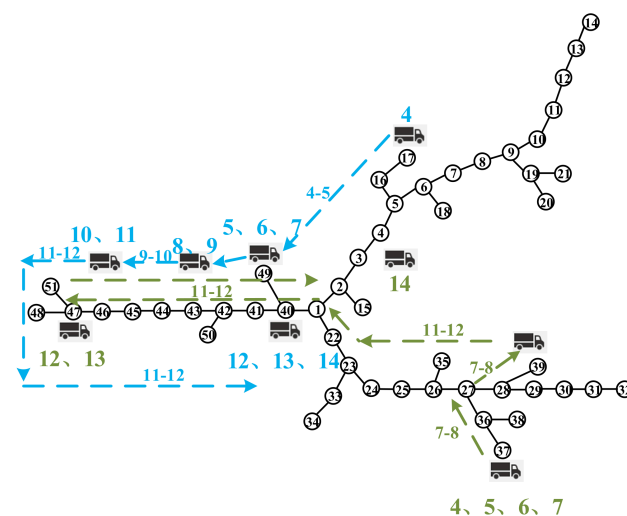


Figure 7. Schematic diagram of dispatch results for the mobile energy storage network.

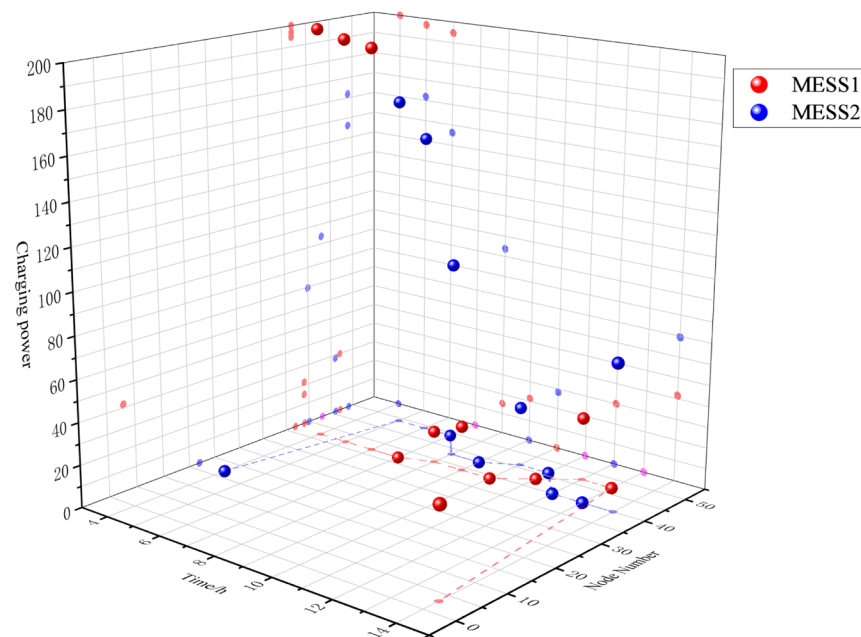


Figure 8. Schematic diagram of power transmission during different time periods for mobile energy storage.

During periods 4–6, corresponding to the initial stage of post-disaster recovery, the system load shedding power exhibits a rapid overall increase. Since a reasonable spatial pre-allocation strategy has been implemented during the pre-disaster stage, the mobile energy storage units can directly participate in power supply without additional relocation after the disaster occurs. Specifically, MESS1 remains connected to node 37 and continuously discharges at a power level close to its rated capacity of 200 kW, while MESS2 is connected to node 49 from period 5 onward, providing 160 kW of power support.

During the early stage of post-disaster restoration (periods 4–6), the system load restoration rate rapidly increased from an initial 41.2% to 68.5% by the end of period 6, with an average recovery of approximately 13.6% of the lost load per period. In this stage, MESS1 cumulatively supplied 587 kWh to loads, and EVCS5 and EVCS15 together fed back 412 kWh through V2G mode; distributed resources contributed approximately 38% of the restoration power in this stage.

Meanwhile, all EVCSs operate in the discharging mode during this stage, and at period 4 they exhibit high-power discharge characteristics. At this time, the EVCSs are transformed in the optimization model from conventional transportation loads into distributed emergency power sources. Their power outputs are effectively superimposed with those of the mobile energy storage systems at the node level, thereby providing direct support for local load restoration.

During the mid-stage of post-disaster recovery (periods 7–10), the overall system load shedding power continues to increase, although the growth rate becomes slower compared with the early stage. From the dispatching results of the mobile energy storage systems, it can be observed that neither MESS participates in discharging at period 7. This is mainly because, after providing high-power support in the previous stage, the energy storage systems must balance state-of-charge (SOC) constraints with subsequent scheduling requirements.

During periods 8–10, the mobile energy storage systems gradually resume participation in power exchange. However, their operational pattern changes from the high-power centralized support observed in the early stage to low-power, cross-node refined compensation. Taking MESS2 as an example, its dispatching range moves between nodes 43 and

46, indicating that the primary role of mobile energy storage during the mid-stage has shifted from enhancing the overall system supply capacity to suppressing the propagation of local load shedding, thereby effectively reducing the growth slope of system load shedding power.

At the same time, the output level of the EVCSs during this stage remains relatively low, and at certain periods the stations even switch between charging and discharging states. This indicates that during the mid-stage recovery process, EVCSs mainly serve as local power balancing and operational constraint regulation resources, rather than as the primary energy sources for load restoration. Nevertheless, because the intensive discharge of EVCSs during the early stage has already significantly reduced the scale of unrecovered loads, the reduction in their output during the mid-stage does not cause any significant adverse impact on the system load shedding level.

During the late stage of post-disaster recovery (periods 11–14), with the gradual parallel operation among distribution network islands, the overall power supply capability of the system is significantly enhanced. During periods 11–13, the mobile energy storage systems are occasionally inactive or in dispatching states. However, no obvious rebound in system load shedding occurs during this stage, indicating that the scheduling decisions implemented in the early- and mid-stages have gradually transformed the distribution network operating mode into one dominated by conventional power supply.

At this stage, mobile energy storage mainly plays the role of maintaining system stability and suppressing potential risks, rather than continuously providing high-power supply. At period 14, the mobile energy storage systems participate in dispatching again and are connected to nodes 47, 3, and 40, providing charging powers of 41 kW and 69 kW to compensate for local loads. Although the load levels at these nodes are relatively small, they may still constrain the overall system restoration in the late recovery stage. The targeted access of mobile energy storage systems at this stage effectively prevents a renewed expansion of load shedding power. Meanwhile, the EVCSs exhibit relatively low output levels in the late stage and only show minor discharging behavior at a few periods, indicating that their role has shifted from emergency power supply in the early stage to auxiliary local regulation resources.

To verify the reasonableness of the earlier assumption, an actual road network topology based on OpenStreetMap data is introduced, generating a non-uniform inter-node travel distance matrix. Comparative analysis shows that under the heterogeneous road network, the total load restored varies by less than 5%, validating the robustness of the proposed method.

5.5. Analysis of Comparative Results

To verify the advantages of the proposed multi-source distributed resource coordinated restoration strategy for improving distribution network resilience, four recovery strategies are designed based on different combinations of distributed energy resources and whether mobile energy storage is pre-allocated, as shown in the table. Simulation studies are conducted for all four strategies.

- Strategy 1 only utilizes distributed generation (DG) resources, including wind turbines, photovoltaic units, and EVCSs, for post-disaster restoration.
- Strategy 2 utilizes pre-allocated mobile energy storage systems together with DG resources.
- Strategy 3 employs mobile energy storage systems without pre-disaster pre-allocation, along with DG resources and EVCSs, for system restoration.
- Strategy 4 corresponds to the proposed method in this paper.

The case study parameters are consistent with those presented in the previous section. Based on the load shedding power of the distribution network during different fault periods, the post-disaster restoration performance curves under different strategies are obtained.

As shown in Figure 9, among the four fault recovery strategies, the proposed road-network–source coordinated restoration scheduling strategy based on multiple distributed resources achieves the lowest load shedding cost, improving system performance by 8.36%, 4.73%, and 1.37% compared with Strategies 1, 2, and 3, respectively.

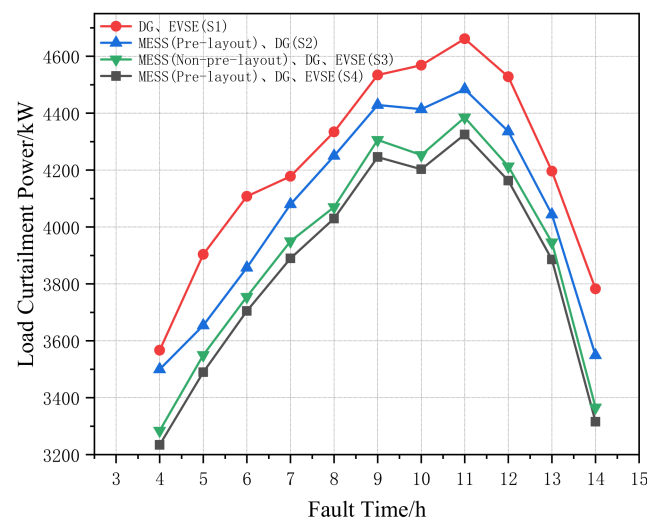


Figure 9. Load reduction power diagram under different strategies.

Since Strategies 3 and 4 both consider the coordinated restoration of all distributed resources and connect them to critical load nodes, their restoration performance is superior to that of Strategies 1 and 2, which only include two types of distributed resources. Furthermore, because the mobile energy storage systems in Strategy 4 are pre-allocated at nodes 17 and 37, which are equivalent to being deployed on the two main trunk paths of the distribution network topology, the energy can be optimally distributed across the network in both temporal and spatial dimensions. Consequently, Strategy 4 achieves the best restoration performance.

6. Conclusions

To effectively mitigate the adverse impacts of typhoon disasters on urban distribution networks, this paper proposes a spatiotemporal coordinated restoration scheduling strategy for “road–network–source” systems based on multiple distributed energy resources. Considering the correlation of distribution network infrastructure failures, a pre-disaster prevention framework centered on resource pre-deployment and proactive network reconfiguration is established, together with a post-disaster restoration system based on the spatiotemporal scheduling of distributed resources. The effectiveness of the proposed strategy is verified through case studies, and the main conclusions are summarized as follows:

- A typhoon spatiotemporal model is constructed based on the urban geographical environment, consisting of a static wind field model, a dynamic evolution model, and a path movement model. Using the typhoon spatiotemporal model as input and based on structural reliability theory, the failure mechanisms of distribution line components are quantitatively analyzed. The results indicate that the angle between wind direction and line orientation, line height, line length, and the geographical environment in

which the line is located are critical variables affecting the time-varying failure rates of distribution line components.

- A distribution network infrastructure failure probability model considering correlation effects is established. The Independent Cascade algorithm is employed to simulate the correlation among poles/towers, while Copula-based joint modeling is used to characterize the dependency between poles/towers and conductors. This approach enables a more comprehensive identification of high-risk branches and branches with high reliability requirements in distribution networks under disaster conditions.
- A two-stage scheduling strategy for disaster prevention and post-disaster restoration is developed for disaster-induced fault scenarios. In the pre-disaster stage, proactive distribution network reconfiguration and pre-deployment of mobile energy storage systems are implemented. In the post-disaster stage, a restoration framework is formed through the spatiotemporal coordinated scheduling of multiple distributed resources, including mobile energy storage systems, electric vehicle charging stations, wind turbines, and photovoltaic units. The proposed strategy effectively reduces the node load shedding cost after disasters and enhances the power supply reliability of critical loads.

The proposed framework was validated on a distribution network in a southeastern coastal city of China, but its methodological core—driving failure propagation through typhoon spatiotemporal evolution and establishing a road-network coupled two-stage restoration dispatch—possesses transferability to other hazard types and geographic regions. For similar wind-induced hazards such as hurricanes and winter storms, only the substitution of the corresponding wind field model and structural fragility parameters is required for reuse. For other extreme events in inland areas (e.g., severe convective storms, ice disasters), the meteorological model and failure correlation structure would need recalibration. For large-scale distribution systems (e.g., with thousands of nodes), the computational efficiency of the current MISOCF formulation may become a bottleneck; future work could adopt Benders decomposition to split fault scenarios into subproblems solved in parallel, or employ solution acceleration strategies based on graph neural network (GNN) surrogate models.

The pre-disaster/post-disaster two-stage dispatch strategy proposed in this paper provides a foundational framework for enhancing the typhoon resilience of distribution networks. Toward practical engineering application, the following three directions merit further investigation: (1) Real-time optimization: Incorporating a model predictive control (MPC) framework to advance post-disaster dispatch from the current hourly rolling to minute-level real-time decision-making, dynamically responding to short-term fluctuations in traffic conditions and renewable generation output. (2) Unified modeling of multi-source uncertainties: Integrating typhoon track forecast errors, PV/wind output uncertainties, and traffic condition uncertainties into a unified Distributionally Robust Optimization (DRO) framework, reducing the conservatism of the robust strategy. (3) Multi-hazard cascading and repair crew coordination: Extending the model to encompass the typhoon–rainstorm–flood cascading hazard chain and its differentiated impacts on underground cables and substations, and incorporating optimal routing and repair time window constraints for repair crews to form a more complete end-to-end post-disaster restoration optimization.

During the course of this study, the continuous multi-period impacts of disasters on the spatiotemporal scheduling of multiple distributed resources in distribution networks have not been considered. Future work will extend the proposed framework to develop a multi-timescale spatiotemporal coordinated restoration strategy for multiple distributed resources under prolonged disaster conditions. Specifically, multi-timescale dispatch can be realized through a three-level hierarchical architecture: pre-disaster day-ahead long-

timescale (24–48 h) robust resource pre-deployment, during-disaster hourly (1 h) rolling optimization dispatch, and post-disaster minute-level (15 min) model predictive control (MPC)-based real-time power correction, achieving seamless transition from long-period prevention to short-period correction through progressive refinement of time scales.

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